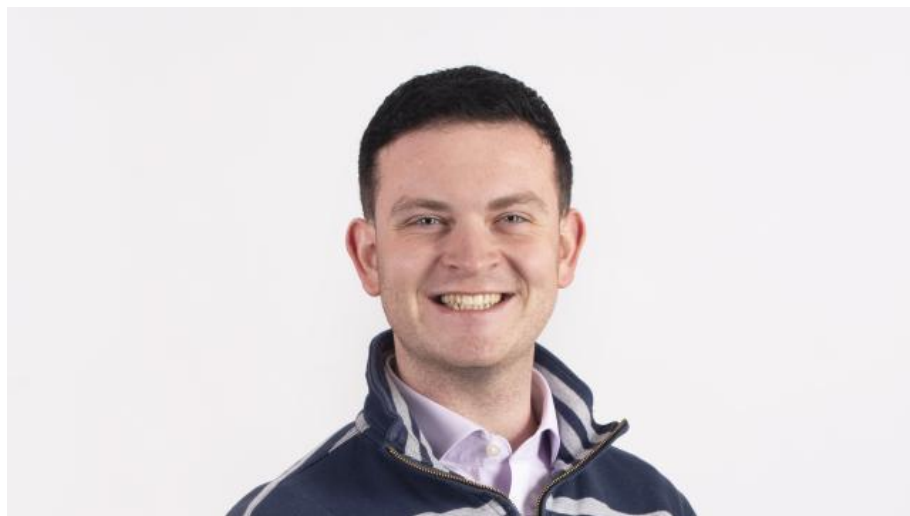




Welcome!



Chris Oldnall



Aybuke Atalay



Descriptive vs Inferential

Descriptive statistics involve summarising and presenting the main features of a dataset, such as measures of central tendency (mean, median, mode) and measures of variability (range, standard deviation), along with identifying the distribution of the data.

Inferential statistics, aims to make predictions or draw conclusions about a population based on a sample of data, using probability theory and statistical inference techniques like hypothesis testing and confidence intervals.





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SESSION A: MEASURES OF CENTRAL TENDENCY AND SPREAD



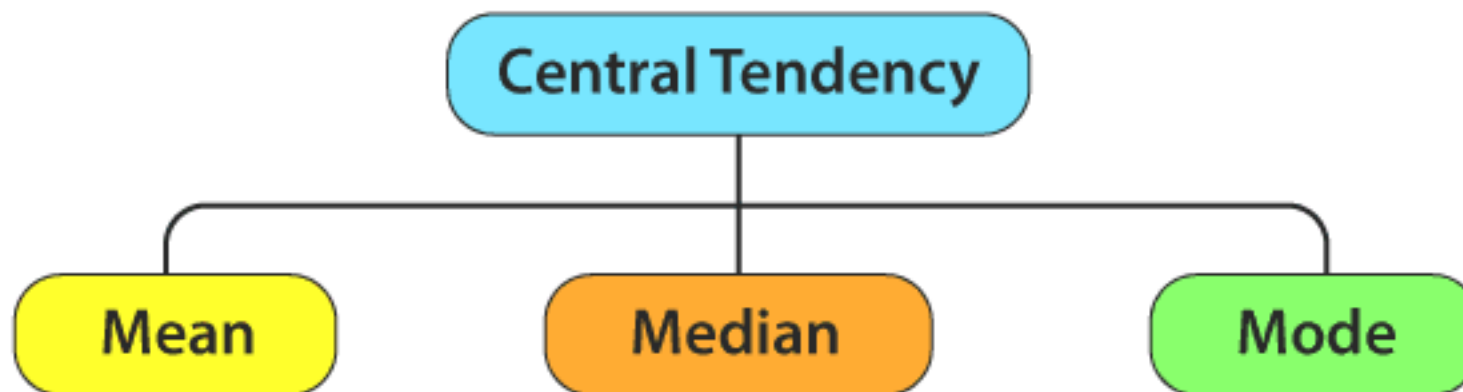
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Why summarise data?

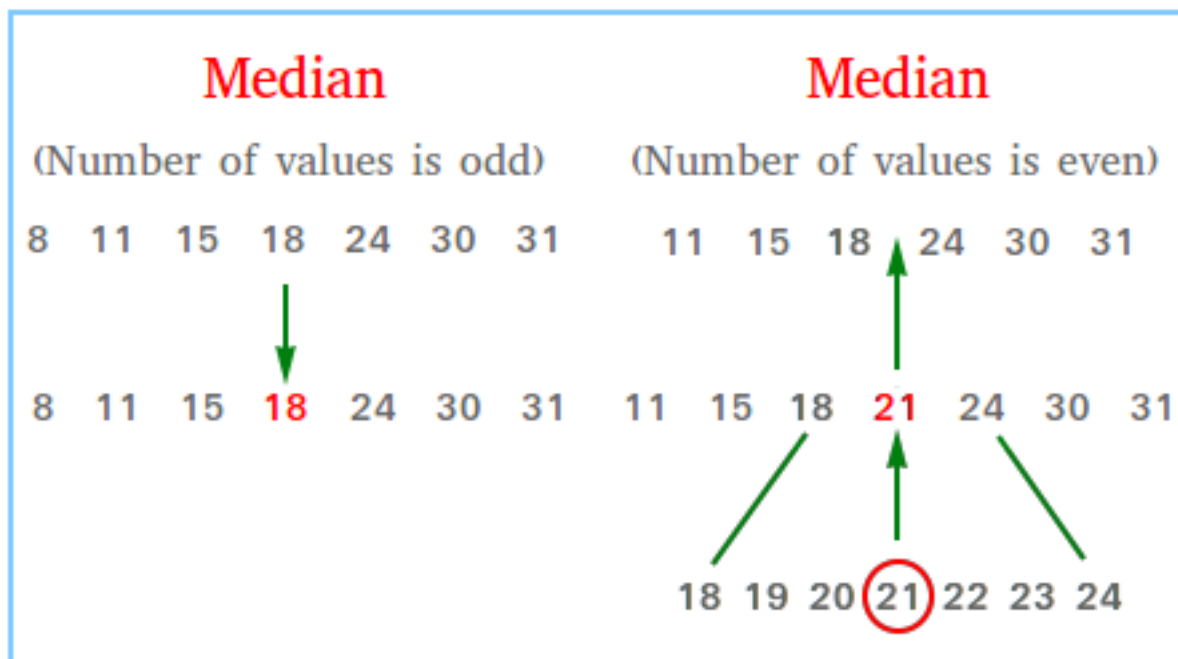
- Real datasets have hundreds, thousands, or millions of values.
- Summary statistics let us extract patterns quickly.
- They help us compare datasets, detect oddities, or check assumptions.
- But: no single number ever tells the full story.



What is a measure of central tendency?



The median.



The mode.

1, 3, 3, 3, 5, 6, 6, 9, 9, 9

There are two modes

3 9

The mean.

Mean Formula

$$\text{Mean} = \frac{\text{Sum of All Data Points}}{\text{Number of Data points}}$$



What is a measure of spread (or dispersion)?



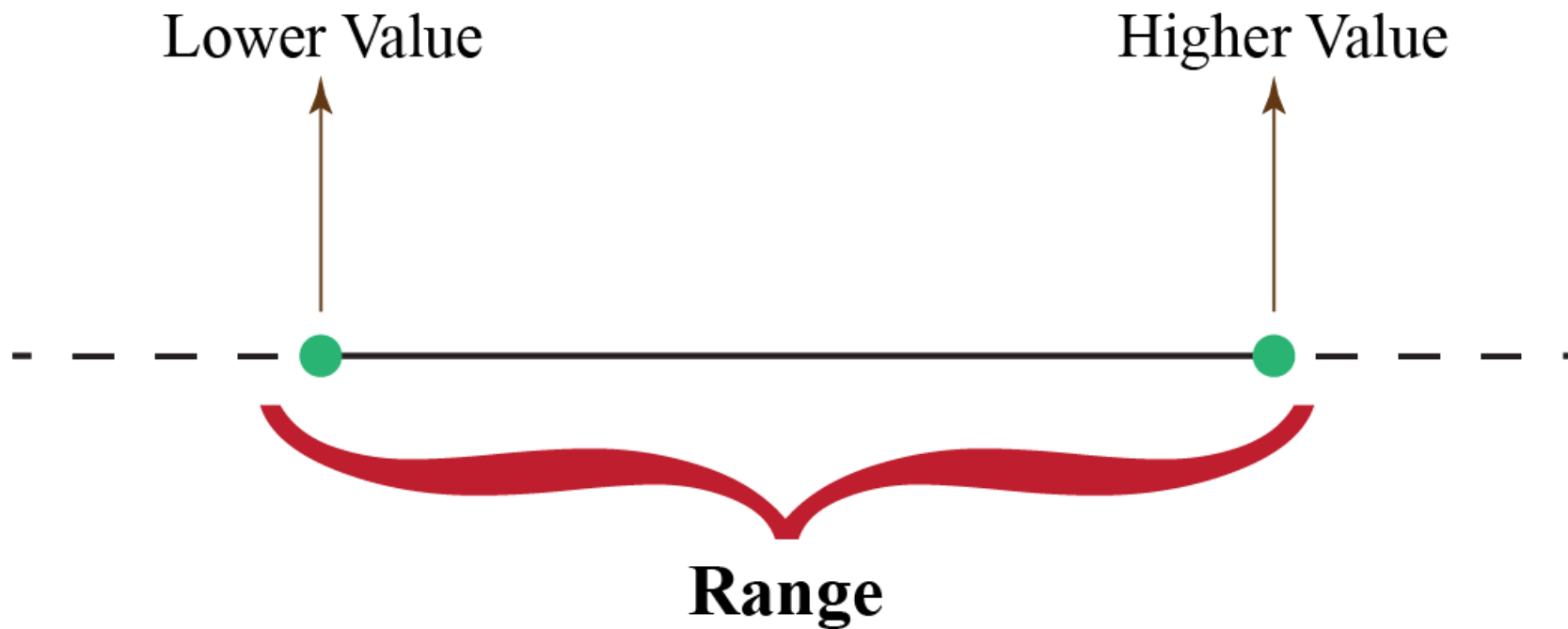
Dataset 1: 5, 5, 5, 5, 5

Dataset 2: 1, 3, 5, 7, 9

Both have mean = 5, but very different spreads.

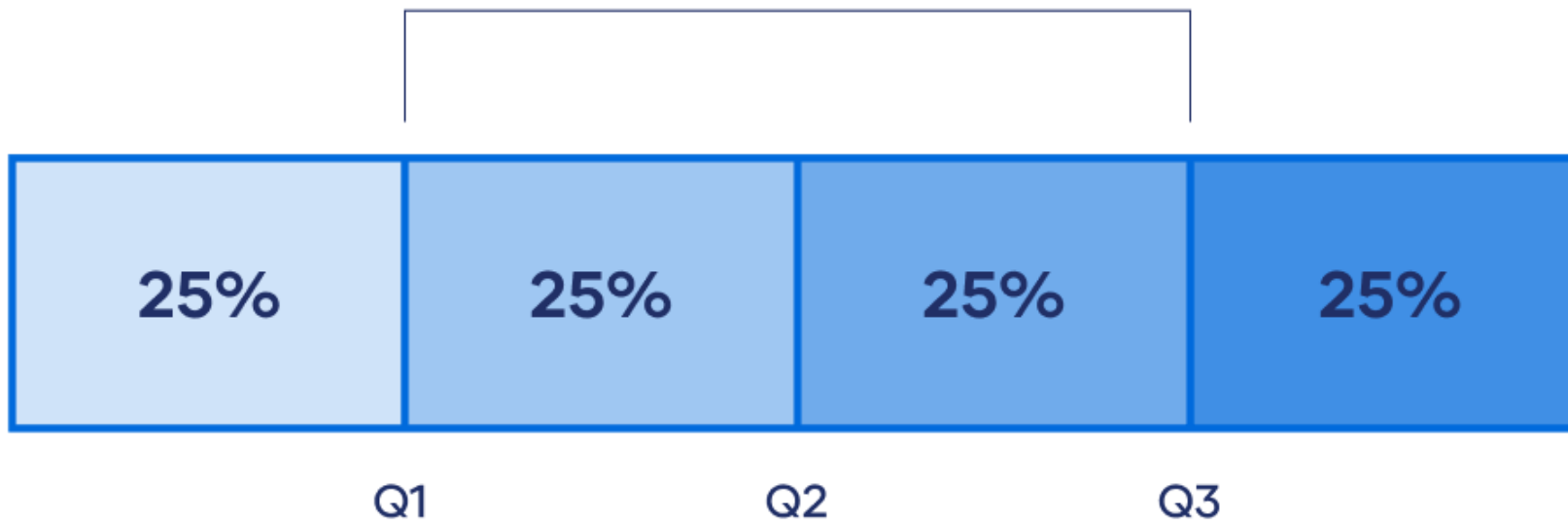


The range.



The IQR.

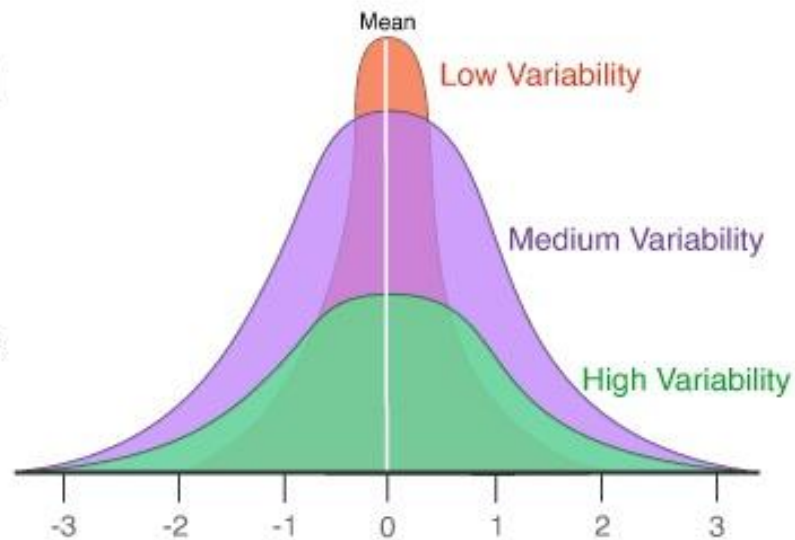
Interquartile range



Standard Deviation.

A statistical method that evaluates the spread or dispersion of a dataset relative to its mean.

Mathematically, it is the average squared deviation of each data point from the mean.



$$\text{Variance} = (\text{Standard Deviation})^2$$

Let's do some programming

Notebook A: What are summary statistics?





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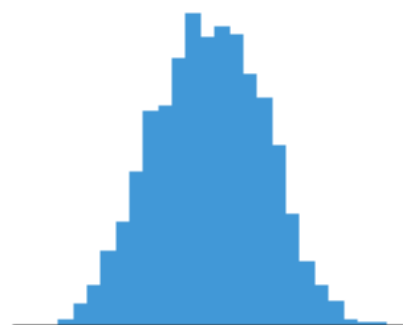
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SESSION B: DISTRIBUTIONS AND VISUALISING STATISTICS

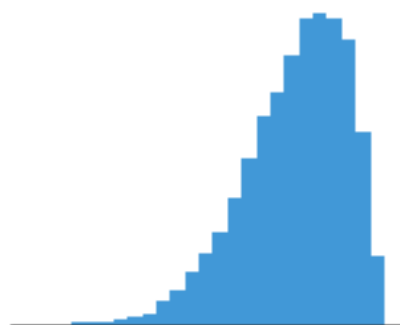


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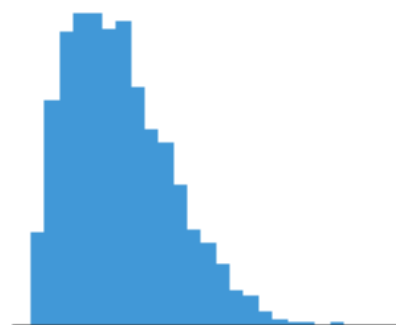
What is a distribution?



symmetric, unimodal



skew left



skew right



uniform



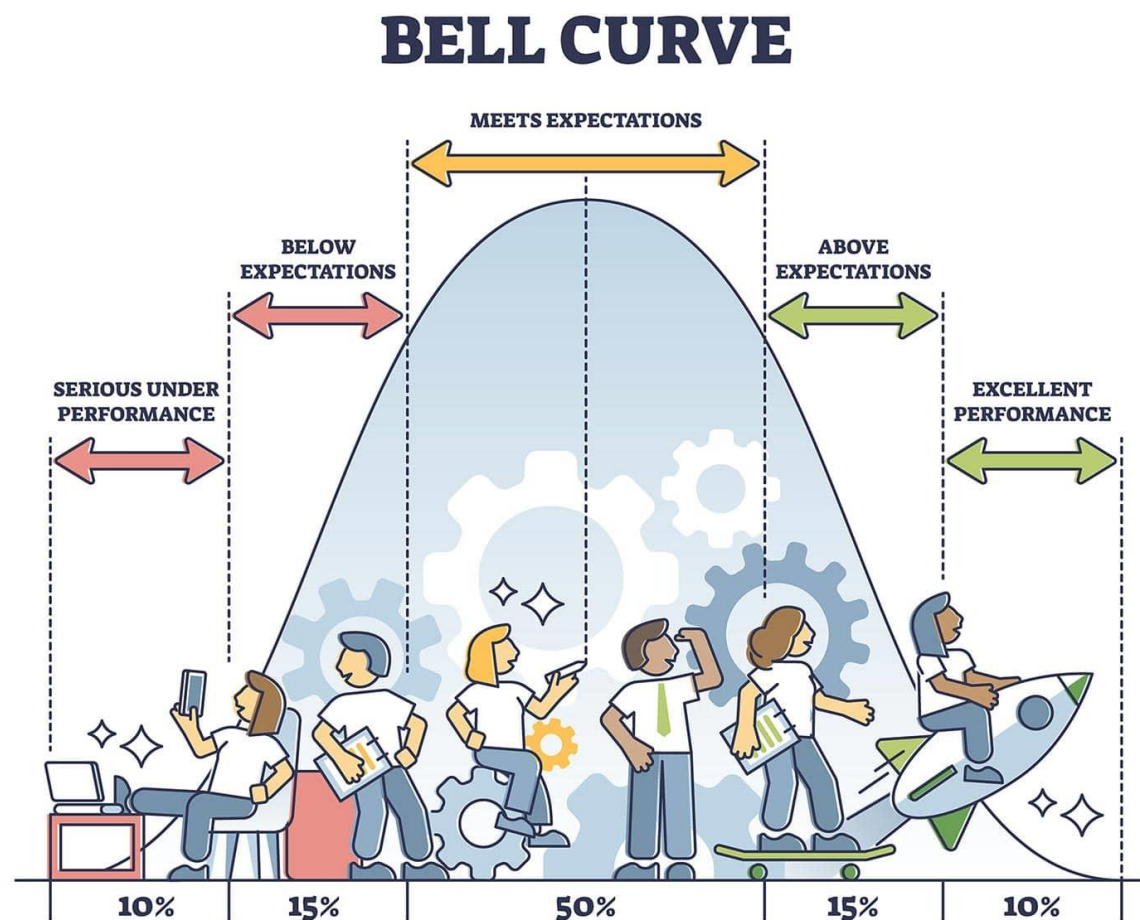
bimodal



multimodal



The normal distribution



How do we identify distributions?

- Histogram: overall shape
- Density plot: smoothed shape; highlights modes
- Facet histograms: compare groups
- Scatterplots: visualise relationships between variables
- Q–Q plot: check for normality





Let's do some programming

Notebook B: Identifying distributions.



Feedback for us...

- We hope you've enjoyed the course as much as we did.
- It is really useful for us to hear your feedback

<https://forms.cloud.microsoft/Pages/ResponsePage.aspx?id=sAafLmkWiUWHiRCgaTTcYbJksOTKjv5Nsdenk5ny0NNUQVZTUK1MS0c3T0JUWENPMIBPMTJLQjJSTC4u>

Should be really quick and only take 5 mins (maximum!)



Thank you from us!



- Please do check out the other training on the CDCS website. We have lots of different things still on offer this semester!

